D4 – Deployment Demonstration and Dashboard

Project: Boréal Marché – Customer Intelligence Platform
Student: Hossam Siyam
Course: 420-977-VA – AI Development Project
Date: September 2026

---

1. Live URLs

Service	URL	Status
API (production)	https://capstone-bkg2.onrender.com	✅ Live
API Documentation (Swagger)	https://capstone-bkg2.onrender.com/docs	✅ Live
Monitoring Dashboard	https://bibe6vpgewedpjfv7emsae.streamlit.app	✅ Live
Source Repository	https://github.com/Abosiyam/Capstone	✅ Public

Note: The API is hosted on Render's free tier, which spins down after 15 minutes of inactivity. The first request after inactivity may take up to 50 seconds to wake the service. This behaviour is documented in the README and the monitoring dashboard.

---

2. API Endpoints

Endpoint	Method	Purpose	Example Response
`/`	GET, HEAD	API information	`{"message":"Boréal Marché Retention API",...}`
`/status`	GET, HEAD	Health check	`{"status":"healthy"}`
`/info`	GET	Model metadata	`{"model":"Logistic_Regression_Calibrated","n_features":25,...}`
`/predict`	POST	Predict one customer	`{"probability":0.78,"decision":"Send coupon"}`
`/predict/batch`	POST	Predict multiple customers	`{"results":[...]}`
`/docs`	GET	Interactive Swagger UI	HTML page

All request and response schemas use **Pydantic v2** with `extra='forbid'` (rejects unknown fields).

---

3. Docker Containerisation

3.1 Dockerfile Design Choices

Choice	Reason
Multi-stage build	Smaller final image – build tools are not included in the runtime image
`python:3.13-slim` **base**	Minimal image (~150 MB) compared to the full Python image (~900 MB)
Non-root user (`appuser`**)**	Security best practice – the container does not run as root
HEALTHCHECK	Docker automatically monitors the API health
Pinned dependencies	`requirements.txt` ensures reproducible builds
`.dockerignore`	Excludes `data/`, `notebooks/`, `venv/` – keeps the image small

3.2 Image Size

Metric	Value
Image name	`boreal-marche-api:latest`
Image ID	`6cef0e440ad3`
Content size	**609 MB** ✅ (under 700 MB "Excellent" target)
Build time	~3–5 minutes

3.3 Build and Run Commands

```
# Build
docker build -t boreal-marche-api .

# Run
docker run -p 8000:8000 boreal-marche-api
```